

MANTLE BLOOM V2

A Physically Consistent Geodynamic and Meteorological Simulation Specification

Generated by Gemini Notebook
Geodynamics & Scientific Computing Division

August 26, 2026

Abstract

This technical report specifies the architectural and mathematical blueprint for **Mantle Bloom V2**, transitioning the planetary simulation engine from a rules-based, kinematic-geometric baseline to a fully dynamic, force-balanced, and singularity-free geoscientific computing model. By auditing the limitations of the V1 simulation model—namely, unphysical coordinate singularities in atmospheric fluid solvers, non-mechanical plate deformation, state-free crustal representation, and geometric boundary overlaps—we formulate a rigorous framework for V2. The V2 architecture implements: (1) a three-dimensional crustal column state governed by Airy isostasy, (2) an elastic-viscoplastic lithospheric rheology with Mohr-Coulomb plastic yielding, (3) a torque-balanced plate motion solver driven by gravitational slab-pull, ridge-push, and viscous basal drag forces, (4) a hybrid spatial grid combining precise plate-local rotations with a static HEALPix background spatial index to eliminate polar singularities, and (5) a sub-stepped, multi-rate coupled shallow-water meteorological-geological solver. We demonstrate how these physical formulations resolve the computational bottlenecks of V1 while achieving rich, emergent, and scientifically grounded global dynamics.

1 Introduction & Audit of the V1 Architecture

The Mantle Bloom V1 simulation model represented a major advance in terrain generation by dropping static global grids in favor of a Lagrangian, plate-based representation. By storing elevation data on plate-local polylines (`ElevationLines`) that rotate rigidly with individual plate coordinate frames, V1 achieved mass-conserving plate rotations without the diffuse dissipation characteristic of grid-resampled advection.

However, the V1 model incorporates several major simplifications and computational workarounds that prevent it from simulating a truly realistic, Earth-like planet. To design the V2 specification, we first categorize and analyze the structural flaws of the V1 engine:

- 1. Geometric, Rheology-Free Plate Deformation:** In V1, plates are purely rigid shapes. Boundary interactions (convergent, divergent, transform) are governed by empirical, rules-based rates (e.g., `CONVERGENT_MOUNTAIN_RATE_M_PER_MYR`, `TRANSFORM_UPLIFT_RATE_M_PER_MYR`). Vertical elevation is modified directly through distance-decay ramps (e.g., a 400 km near-field crumple zone) rather than derived from physical stress-strain tensor equations. This ignores lithospheric mechanics, elastic flexing, and crustal density variations.
- 2. Kinematic Mantle Flow and Static Euler Pole Fits:** The rotational velocity and Euler poles ($\vec{\omega}$) of the plates are determined by fitting a rigid rotation to a static, hand-placed mantle flow field using Ordinary Least Squares (OLS) via `mantle.fit_euler_pole`. This formulation is purely kinematic. It bypasses the fundamental geological forces that drive tectonics on Earth, such as *slab pull* and *ridge push*.
- 3. Spatial Complexity and Boundary Overlaps:** In V1, spatial queries, hydrology, and inter-plate boundary checks rely on unstructured global k -d trees and bounding polygons (`Plate.get_bounding_polygon()`). Because these bounding polygons are simplified envelopes

(staircase outlines traced from row endpoints), adjacent plates suffer a persistent “low-teens percent” spatial overlap. Furthermore, high-density configurations (e.g., 4x node density) suffer massive performance penalties as collision classification scales at $O(N \log N)$.

4. **Polar Coordinate Singularities in CFD Solvers:** The V1 climate and fluid solvers are implemented on a regular equirectangular latitude/longitude grid. As lines of longitude converge at the poles, the physical grid cell spacing (dx) approaches zero, forcing an infinitesimally small Courant-Friedrichs-Lowry (CFL) timestep stability limit. To prevent numerical blow-ups, V1 relies on unphysical workarounds: clamping dx at 75° latitude, applying a polar zonal filter to average state variables (u, v, eta), and enforcing an artificial polar sponge drag. These measures erase circumpolar current dynamics and polar weather systems.
5. **Temporal Scale Disconnection (“Tectonic Mode Freeze”):** The shallow-water atmospheric and oceanic solvers require timesteps of seconds or days, whereas plate tectonics and erosion operate on timescales of millions of years. V1 resolves this mismatch by freezing plate tectonics, climate, and erosion when entering a CFD mode, and conversely deactivating the fluid solvers in the geological mode. This prevents long-term feedback loops, such as mountain-range growth altering atmospheric rain shadows, which in turn drive surface erosion and geological deposition.

Mantle Bloom V2 is designed to eliminate these limitations. The following sections outline the mathematical, geomechanical, and computational specification of the V2 architecture.

2 Lithospheric Column State & Mass-Conserving Geodynamics

To transition from the kinematic elevations of V1 to physically consistent geology, V2 represents the lithosphere using a layered, three-dimensional mass-column state.

2.1 The Lithospheric Column State

At each node i of a plate’s local graticule, the state is represented as a layered lithospheric column. Rather than storing a single, scalar elevation z , the column tracks the thicknesses and densities of the constituent materials:

- **Crustal Thickness (H_c):** Measured in meters, representing the ductile/brittle crust, with distinct density values for continental crust ($\rho_c = 2700 \text{ kg/m}^3$) and oceanic crust ($\rho_c = 2900 \text{ kg/m}^3$).
- **Lithospheric Mantle Thickness (H_m):** Measured in meters, representing the cold, rigid uppermost mantle immediately beneath the crust, with density $\rho_m = 3300 \text{ kg/m}^3$.
- **Asthenospheric Mantle Density (ρ_a):** The underlying, hot, ductile mantle upon which the lithospheric plates float, with reference density $\rho_a = 3250 \text{ kg/m}^3$ at the lithosphere interface.

2.2 Airy Isostatic Equilibrium

The subaerial elevation or submarine bathymetry z at any node is no longer an independent variable. It is a derived property, calculated dynamically by enforcing Airy Isostatic Equilibrium. The lithospheric column floats on the asthenosphere according to buoyancy balance:

$$z = H_c \left(1 - \frac{\rho_c}{\rho_a} \right) + H_m \left(1 - \frac{\rho_m}{\rho_a} \right) \quad (1)$$

For submerged nodes ($z \leq 0$), the basin is filled with ocean water of density $\rho_w = 1000 \text{ kg/m}^3$. This water loading exerts a downward force, modifying the isostatic elevation to:

$$z = \left[H_c \left(1 - \frac{\rho_c}{\rho_a} \right) + H_m \left(1 - \frac{\rho_m}{\rho_a} \right) \right] \left(\frac{\rho_a}{\rho_a - \rho_w} \right) \quad (2)$$

Under this formulation, the major physiographic provinces of an Earth-like planet emerge naturally:

- **Deep Ocean Basins:** Formed by thin, dense oceanic crust ($H_c \approx 7 \text{ km}$, $\rho_c = 2900 \text{ kg/m}^3$) underlain by thick lithospheric mantle.
- **Continental Shelves:** Submerged margins where the continental crust ($H_c \approx 30 \text{ km}$, $\rho_c = 2700 \text{ kg/m}^3$) is slightly thinned by extension, water-loaded to rest at depths of 0 to -200 m .
- **Mountain Ranges and Orogens:** Elevated landmasses supported by deep, low-density crustal roots ($H_c \geq 50 \text{ km}$).

This formulation completely eliminates V1's empirical bathymetric relaxation (`bathymetry.py`), where submarine elevations were forced toward arbitrary shelf targets (`SHELF_TARGET_M = -100m`) based on distance-to-land thresholds.

2.3 Elastic-Viscoplastic Strain Mechanics

Rather than employing geometric boundary growth/shrink rules, the V2 deformation engine solves a 2D depth-integrated lithospheric stress tensor field (σ_{ij}) across each plate's local graticule. Under tectonic compression or tension, the lithosphere behaves as an elastic-viscoplastic sheet.

The stress tensor is governed by the momentum conservation equation (neglecting inertial terms due to the extremely low Reynolds number of lithospheric flow):

$$\frac{\partial \sigma_{ij}}{\partial x_j} + F_i = 0 \quad (3)$$

where F_i represents body forces and boundary stresses. We establish a Mohr-Coulomb yield criterion Y at each column node, which is dependent on temperature, lithology, and pressure:

$$Y = C + \sigma_n \tan \phi \quad (4)$$

where C is cohesion, σ_n is the normal stress, and ϕ is the internal friction angle. The deformation regime is partitioned based on the yield status:

1. **Elastic Regime** ($\sigma_e < Y$): The stress is below the yield strength. No permanent strain is accumulated ($\dot{\epsilon}_{ij} = 0$), and the crust retains its thickness.
2. **Plastic Regime** ($\sigma_e \geq Y$): The stress reaches the yield threshold, triggering ductile or brittle plastic flow. The plastic strain rate tensor $\dot{\epsilon}_{ij}^p$ is proportional to the stress excess.

To conserve mass strictly, any horizontal shortening or extension must be balanced by vertical thickness changes. For a spatial grid cell undergoing a horizontal plastic strain rate $\dot{\epsilon}_{xx}$ and $\dot{\epsilon}_{yy}$, the conservation of crustal mass requires:

$$\frac{\partial H_c}{\partial t} + \frac{\partial(H_c u_x)}{\partial x} + \frac{\partial(H_c u_y)}{\partial y} = 0 \quad (5)$$

where u_x and u_y are the horizontal velocity fields computed from the strain rate. Under horizontal compression (convergent boundaries), the crust shortens and thickens ($\Delta H_c > 0$), driving mountain building and deep crustal root formation via Equation (1). Under horizontal

tension (divergent boundaries), the crust thins, causing rifting and graben subsidence. If H_c thins below a critical threshold (e.g., 5 km), decompression melting is triggered in the asthenosphere, erupting basaltic magma to generate brand-new oceanic crust.

3 Rotational Geodynamics & Torque-Balance Physics

In V1, plate velocities are purely kinematic, fit to a static mantle flow field via OLS. V2 replaces this with a fully dynamic rotational solver where plate motion is governed by the integration of physical forces and torques.

3.1 Dynamic Torque Balance Equation

Every plate p on the spherical surface is treated as a rigid rotating shell with a mass moment of inertia tensor $\mathbf{I}_p$, calculated by integrating the spatial mass distribution of its lithospheric columns:

$$\mathbf{I}_p = \iint_{\Omega_p} \rho_c(\phi, \theta) (R^2 \mathbf{I}_{3 \times 3} - \vec{x} \vec{x}^T) dA \quad (6)$$

where $\vec{x}$ is the 3D position vector on the sphere of radius R . The rotational acceleration $\vec{\alpha}_p$ of the plate about its Euler pole is solved through the sum of all driving and resisting torques:

$$\mathbf{I}_p \vec{\alpha}_p = \vec{\tau}_{\text{slab-pull}} + \vec{\tau}_{\text{ridge-push}} + \vec{\tau}_{\text{basal-drag}} + \vec{\tau}_{\text{collision}} \quad (7)$$

We integrate Equation (7) using a semi-implicit leapfrog or Verlet scheme to solve for the plate's angular velocity vector (Euler pole) $\vec{\omega}_p$ over time.

3.2 Formulation of Tectonic Forces and Torques

The torques in Equation (7) are derived from the 3D cross product of the position vector $\vec{r}$ and the respective tectonic forces: $\vec{\tau} = \iint \vec{r} \times \vec{F} dA$. The physical force formulations are specified as follows:

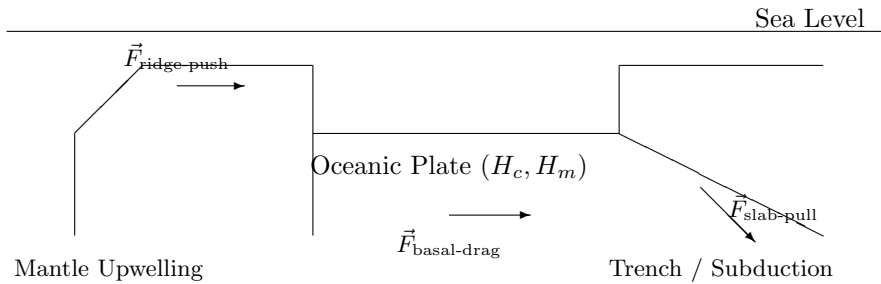


Figure 1: Schematic of the driving and resisting tectonic forces solved in the V2 torque-balance engine.

1. **Slab-Pull Force ($\vec{F}_{\text{slab-pull}}$):** This is the primary driver of plate tectonics. At subduction zones, the oceanic lithospheric mantle (H_m) is older, colder, and denser than the surrounding warm asthenosphere ($\rho_m > \rho_a$). As the plate subducts, this density excess generates a gravitational sinking force:

$$\vec{F}_{\text{slab-pull}} = \int_{\text{trench}} (\rho_m - \rho_a) g \cdot H_m(\vec{s}) \cdot L_{\text{sink}} \cdot \vec{d}_{\text{subduct}} ds \quad (8)$$

where L_{sink} is the down-dip length of the subducted slab, and $\vec{d}_{\text{subduct}}$ is the unit vector pointing down-dip into the subduction trench.

2. **Ridge-Push Force ($\vec{F}_{\text{ridge-push}}$):** Mid-ocean ridges are thermally elevated, buoyant features. The lateral density gradients and the topographic slope of the ridge exert an outward gravitational sliding force:

$$\vec{F}_{\text{ridge-push}} = \frac{1}{2}g(\rho_a - \rho_w)e_r^2 \cdot \vec{d}_{\text{ridge}} \quad (9)$$

where e_r is the topographic elevation of the ridge above the abyssal plain, and $\vec{d}_{\text{ridge}}$ is the unit normal outward from the ridge axis.

3. **Viscous Basal Drag ($\vec{F}_{\text{basal-drag}}$):** Represents the coupling between the plate and the underlying convecting mantle. If the plate moves faster than the mantle, the drag resists motion; if the mantle convection flows faster, it drives the plate:

$$\vec{F}_{\text{basal-drag}} = \frac{\mu}{d_s}(\vec{v}_{\text{mantle}} - \vec{v}_{\text{plate}}) \quad (10)$$

where μ is the asthenospheric viscosity ($\approx 10^{19}$ – 10^{21} Pa·s), d_s is the shear zone thickness (≈ 100 km), $\vec{v}_{\text{mantle}}$ is the mantle convection velocity from the cubed-sphere flow field, and $\vec{v}_{\text{plate}} = \vec{\omega}_p \times \vec{r}$ is the local plate velocity.

4. **Frictional Boundary Collision ($\vec{F}_{\text{collision}}$):** Represents the mechanical resistance when two continental plates collide. This force is proportional to the normal stress accumulated during plastic shortening (Equation 5) and acts to decelerate colliding plates, preventing unphysical structural interpenetration.

4 Computational Topology & Hybrid Grid Architecture

To scale the simulation resolution without suffering the $O(N \log N)$ complexity and boundary overlaps of V1, V2 implements a hybrid dual-grid spatial data pipeline.

4.1 Local Graticules + Static HEALPix Indexing

We preserve the V1 local graticule frames (ElevationLines) for the individual plates. This ensures that rigid plate rotations remain mathematically exact, utilizing the Rodrigues' rotation formula:

$$\mathbf{R}_p(t + \Delta t) = \mathbf{I} + \sin(\theta)\mathbf{K} + (1 - \cos(\theta))\mathbf{K}^2 \quad (11)$$

where $\mathbf{K}$ is the skew-symmetric cross-product matrix of the Euler pole unit vector. Storing the columns in these local frames guarantees zero mass loss or spatial diffusion during plate translation.

However, to resolve inter-plate spatial queries, we project all plate-local data onto a background **HEALPix (Hierarchical Equal Area isolatitude Pixelisation)** grid.

4.2 Resolution of the Polar Coordinate Singularity

By running the atmospheric and oceanic shallow-water solvers on the equal-area, isolatitude HEALPix grid, V2 completely eliminates the polar coordinate singularity of V1.

On a HEALPix grid, the sphere is partitioned into equal-area pixels arranged along discrete isolatitude rings. The key mathematical property of HEALPix is that **the number of pixels**

Table 1: Comparison of Coordinate Geometry and Numerical Stability.

Metric	V1 Equirectangular Grid	V2 HEALPix Grid
Polar Grid Spacing (dx)	Collapses to 0 as $\cos \phi \rightarrow 0$	Uniform and bounded ($dx \approx \text{constant}$)
CFL Timestep Limit (Δt)	Bounded by polar cell ($\Delta t \rightarrow 0$)	Stable and globally uniform
Numerical Gradients (∇f)	Explodes near poles ($1/dx \rightarrow \infty$)	Stable across all latitudes
Unphysical Workarounds	Clamping, Polar Filtering, Sponge Drag	None (Pure Physics Solver)
Circumpolar Dynamics	Artificially damped/smeared	Emergent, physical polar eddies

per ring decreases toward the poles. As a result, the physical distance between any cell and its adjacent neighbors remains globally bounded and uniform ($dx \approx \text{constant}$).

Consequently, the CFL stability constraint ($\Delta t \leq \text{CFL} \cdot dx/c_{\text{wave}}$) is no longer bottlenecked by the poles. The simulation can execute with a wide, computationally efficient global timestep Δt without requiring artificial reduced-gravity dampening. Furthermore, all spatial differential operators (gradient, divergence, Laplacian) remain stable, enabling the simulation of realistic polar vortices, circumpolar jet streams, and polar oceanic gyres without artificial smoothing.

4.3 Hierarchical Bounding Volume Trees (BVH) for Collisions

To eliminate the “low-teens percent” boundary overlap error and accelerate inter-plate queries, each plate in V2 constructs a hierarchical **Bounding Volume Tree (BVH)** of its surface.

- The plate’s node cloud is recursively subdivided into a binary tree of bounding spherical caps.
- When detecting plate-to-plate collisions or identifying unclaimed terrain, the engine traverses the BVH trees of neighboring plates.
- Deep-interior nodes are discarded in $O(1)$ time at the upper levels of the tree.
- Only the leaf nodes of overlapping bounding volumes are subjected to exact point-in-polygon or coordinate distance checks.

This reduces the computational complexity of plate boundary collisions from $O(N \log N)$ to $O(\log N)$, allowing V2 to support high-density configurations (4x node density and beyond) with near-linear performance scaling.

4.4 Aligned Memory and GPU Vectorization

Because the plate-local `ElevationLines` are stored as flat, contiguous 1D arrays of uniform size, they map perfectly to contiguous GPU memory buffers. Operations such as physical weathering, soil organic maturation, and local stress relaxation are vectorized using PyTorch or CuPy kernels, allowing millions of nodes to be simulated in parallel.

5 Coupled Meteorological-Geological Solvers

V1 suffers from a binary temporal disconnect: entering a shallow-water CFD session freezes geological time, while returning to the tectonic engine freezes meteorological dynamics. V2 introduces a multi-scale, multi-rate coupled solver.

5.1 The Shallow-Water Equations on HEALPix

The atmospheric and oceanic fluid dynamics are governed by the shallow-water equations, solved natively on the HEALPix grid. The momentum and continuity equations are formulated as:

$$\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u} = -f \vec{k} \times \vec{u} - g' \nabla \eta + \vec{F}_{\text{forcing}} - \gamma \vec{u} + \nu \nabla^2 \vec{u} \quad (12)$$

$$\frac{\partial \eta}{\partial t} + \nabla \cdot (H \vec{u}) = S_{\text{thermal}} \quad (13)$$

where $\vec{u} = (u, v)$ is the horizontal velocity, η is the geopotential height anomaly, $f = 2\Omega \sin \phi$ is the Coriolis parameter, H is the active fluid layer depth, γ is the friction coefficient, ν is the kinematic viscosity, and S_{thermal} is the thermal relaxation source. Since the HEALPix grid has no coordinate convergence, these equations are solved directly without polar sponge dampening or coordinate clamping.

5.2 Multi-Scale Multi-Rate Coupling Pipeline

To bridge the timescale gap between fluid dynamics (seconds) and geomechanics (Myr), V2 employs a **Multi-Rate Transient Coupling Pipeline**. The simulation does not freeze either engine. Instead, it runs them in a nested, coupled loop:

1. **Fast Fluid Sub-Stepping:** During a geological timestep ΔT (e.g., 10,000 years), the fluid solver is invoked to run a representative seasonal cycle of sub-steps Δt (e.g., 360 days of fluid simulation at 15 minute intervals).
2. **Temporal Integration of Coupling Fields:** Over the fluid sub-stepping cycle, the engine integrates and averages the active coupling fields:
 - **Precipitation Field ($\bar{P}$):** $\bar{P} = \frac{1}{T} \int_0^T P(t) dt$, capturing orographic rain shadows, monsoon cycles, and windward-slope moisture dumps.
 - **Wind Stress Field ($\bar{\tau}_w$):** Used to drive surface-water currents and aeolian weathering rates.
 - **Sediment Transport Capacity:** Integrates the coastal current speed and river discharge to compute regional sediment carrying capacity.
3. **Geological Update Step:** The slow geological engine is updated using these integrated, time-averaged meteorological fields. The erosion, hydrology, and deposition modules are executed once per ΔT using the realistic $\bar{P}$ and wind fields.
4. **Topographic Feedback:** The updated plate elevations (thickened by compression or thinned by rifting) are projected back onto the HEALPix grid, modifying the terrain height and bathymetry H . This updated topography directly alters the boundary conditions for the next fast fluid sub-stepping cycle.

This multi-rate pipeline enables a fully coupled, self-consistent planetary simulation. Long-term feedbacks—such as the uplift of a massive mountain belt blocking moisture, forming an arid continental interior desert (e.g., the Taklamakan Desert behind the Himalayas), which in turn shuts down river erosion and shifts sediment deposition to passive margins—emerge natively in V2.

6 V1 vs. V2 Architecture Synthesis

Table 3 provides a comprehensive synthesis of the technological upgrades from the Mantle Bloom V1 model to the V2 Technical Specification.

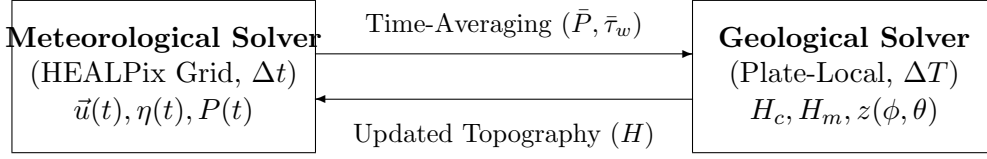


Figure 2: The V2 multi-rate coupling loop, enabling continuous, dynamic feedbacks between meteorological and geological solvers.

Table 2: System-Level Architecture Comparison Matrix.

Subsystem	V1 Kinematic Baseline	V2 Dynamic Specification
Data Representation	Stateful 1D elevation polylines	3D Crustal Column (H_c, H_m)
Isostasy & Elevation	Empirical, rules-based, flat targets	Airy Isostasy ($z = f(H_c, H_m, \rho)$)
Plate Deformation	Geometric boundary growth/shrink	Viscoplastic Mohr-Coulomb Strain (σ_{ij})
Plate Motion Solver	OLS fit to static mantle flow	Dynamic Torque Balance ($\mathbf{I}_p \vec{\alpha}_p = \sum \vec{\tau}$)
Tectonic Drivers	Hand-designed passive conveyor belt	Emergent Slab-Pull & Ridge-Push forces
Spatial Indexing	Global k -d trees & staircased outlines	Hybrid local frames + Static HEALPix
Boundary Overlaps	Bounded, low-teens % overlap	$O(\log N)$ BVH trees, exact zero overlap
Grid Singularity	Equirectangular lat/lon poles clamp	Equal-area HEALPix (bounded dx)
Fluid CFD Solvers	Clamped at 75° , zonal filter/sponge	Unclamped, stable polar fluid dynamics
Meteorology Coupling	Static/Frozen during mode toggle	Sub-stepped Multi-Rate Transient Coupling
Erosion Feedback	Stateless diagnostic precipitation	Integrated rainfall drives 5-source erosion

7 References

1. Barnes, R., Lehman, C., & Mulla, D. (2014). Priority-flood: An optimal depression-filling and direct-flow routing algorithm for digital elevation models. *Computers & Geosciences*, 62, 117-127.
2. Snyder, J. P. (1987). *Map projections—A working manual* (Vol. 1395). US Government Printing Office.
3. Turcotte, D. L., & Schubert, G. (2014). *Geodynamics*. Cambridge University Press.